Catalogue of proofs

This list shows where to find the proofs in the notes. It does not fix a universal number of « required proofs": the groupings vary depending on the presentation.

Proven result	Proof
Compare two sequences at infinity	p. 23
Increasing and not bounded above	p. 23
Bernoulli and geometrical limits	p. 23
Limits of the exponential	p. 33
Powers and the exponential	p. 33
Convexity and tangents	p. 53
Limit of x ln(x) at zero	p. 76
Antiderivative given by an area, increasing case	p. 107
Integration by parts	p. 107
Pascal by counting	p. 43
Pascal by calculation	p. 43
Number of subsets: double counting	p. 43
Minimum distance to the plane	p. 86
Cartesian equation of a plane	p. 131
Probability of k successes	p. 65
Binomial expectation and variance	p. 119
Chebyshev on a finite sample space	p. 119
Derivative of the logarithm (differentiability accepted)	p. 76
Two antiderivatives differ by a constant	p. 97
Solutions of $y' = ay$	p. 98
Concentration of the mean, then the law of large numbers	p. 119